

Software Engineering

BCA — Semester IV — 2021 — Answer Sheet

Subject Code: CACS253

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group C Attempt any TWO questions[2x10=20]

2. What is software engineering? Explain the challenges facing software engineering.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What is extreme programming? Explain how the principles underlying extreme programming lead to the accelerated development of software.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Define quality assurance? Under what circumstances would you recommend the use of staged representation of the CMM?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is architectural design? Explain layered model of software architecture with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Differentiate between verification and validation. Explain why software inspection is an effective technique for discovering errors in software?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is software reengineering? List its advantages. Explain source code translation and reverse engineering approach in brief.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What do you mean by software quality management? Explain software quality management activities in brief.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is software process model? List different software process model. Explain how both the waterfall model of the software process and prototyping model can be accommodated in the spiral process model.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What do you mean by software requirement? Differentiate between functional requirement and non-functional requirement. List the functional requirements of online examination system and represent it in use case diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. a) What is risk management? Explain risk management process in software engineering with block diagram. b) List good programming practices. Explain CASE and its type with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.